

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	5
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	5
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	5
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	5
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	5
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	5

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	5
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	5
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	5
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	5

Code of Conduct:

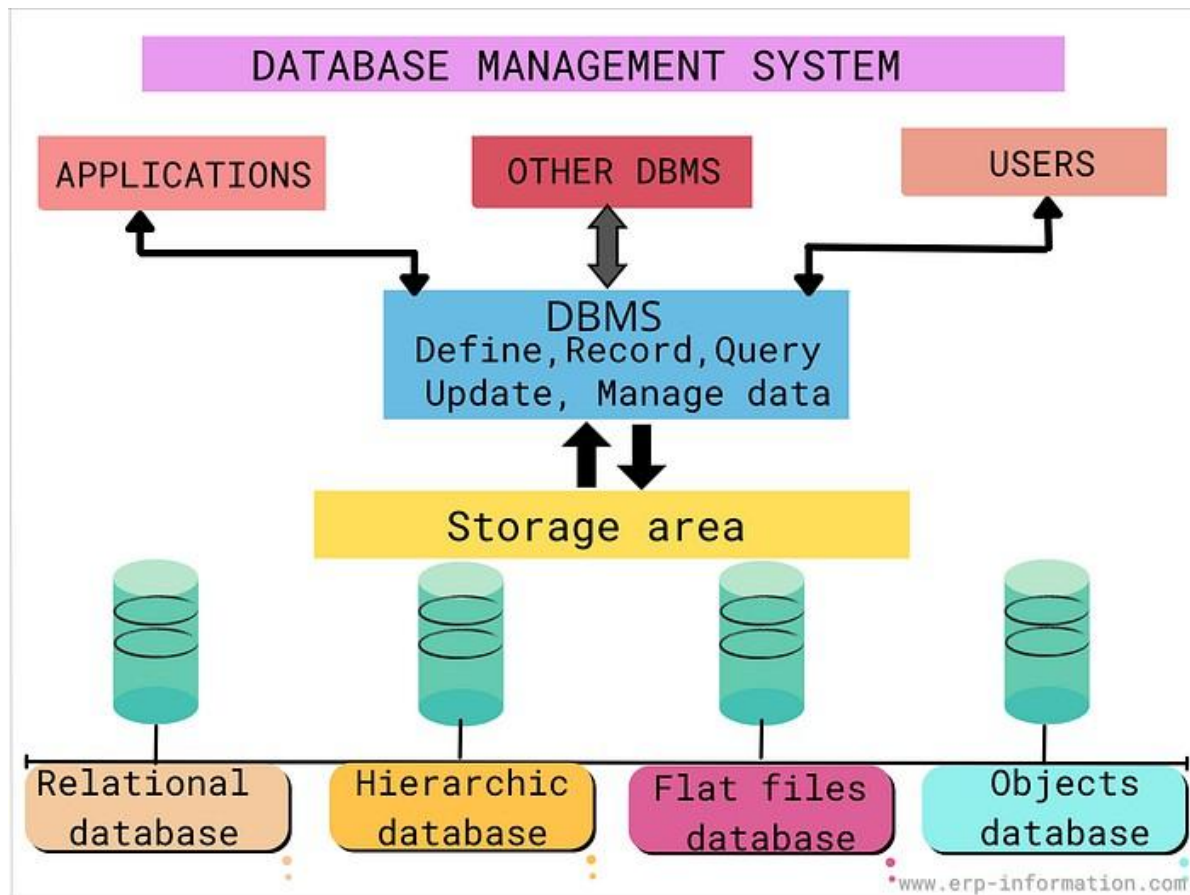
I E . B . A s h w i n i / 1 9 2 5 1 1 4 2 2 _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ashwini

RUBRICS FOR CALCULATION:

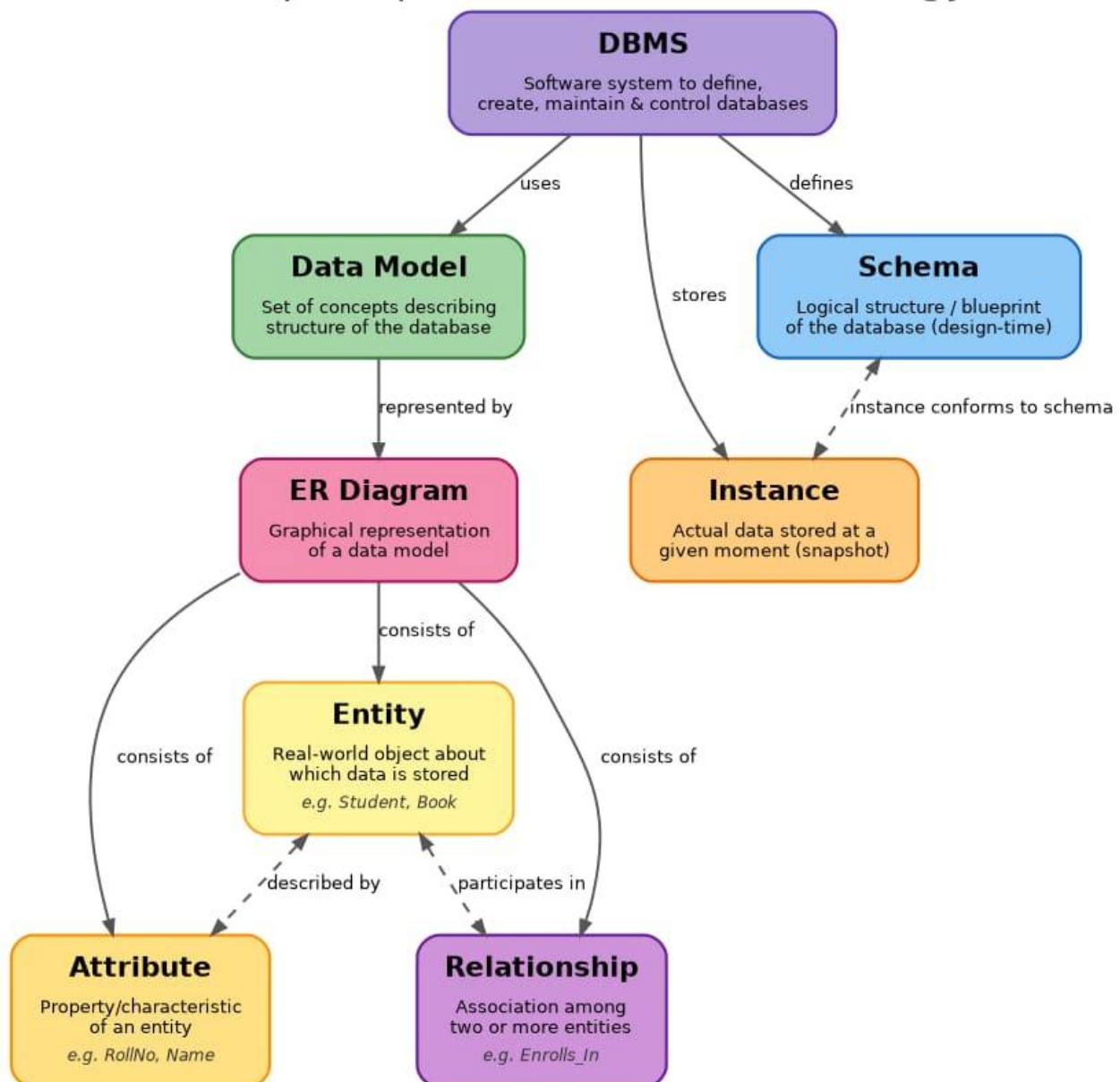
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand



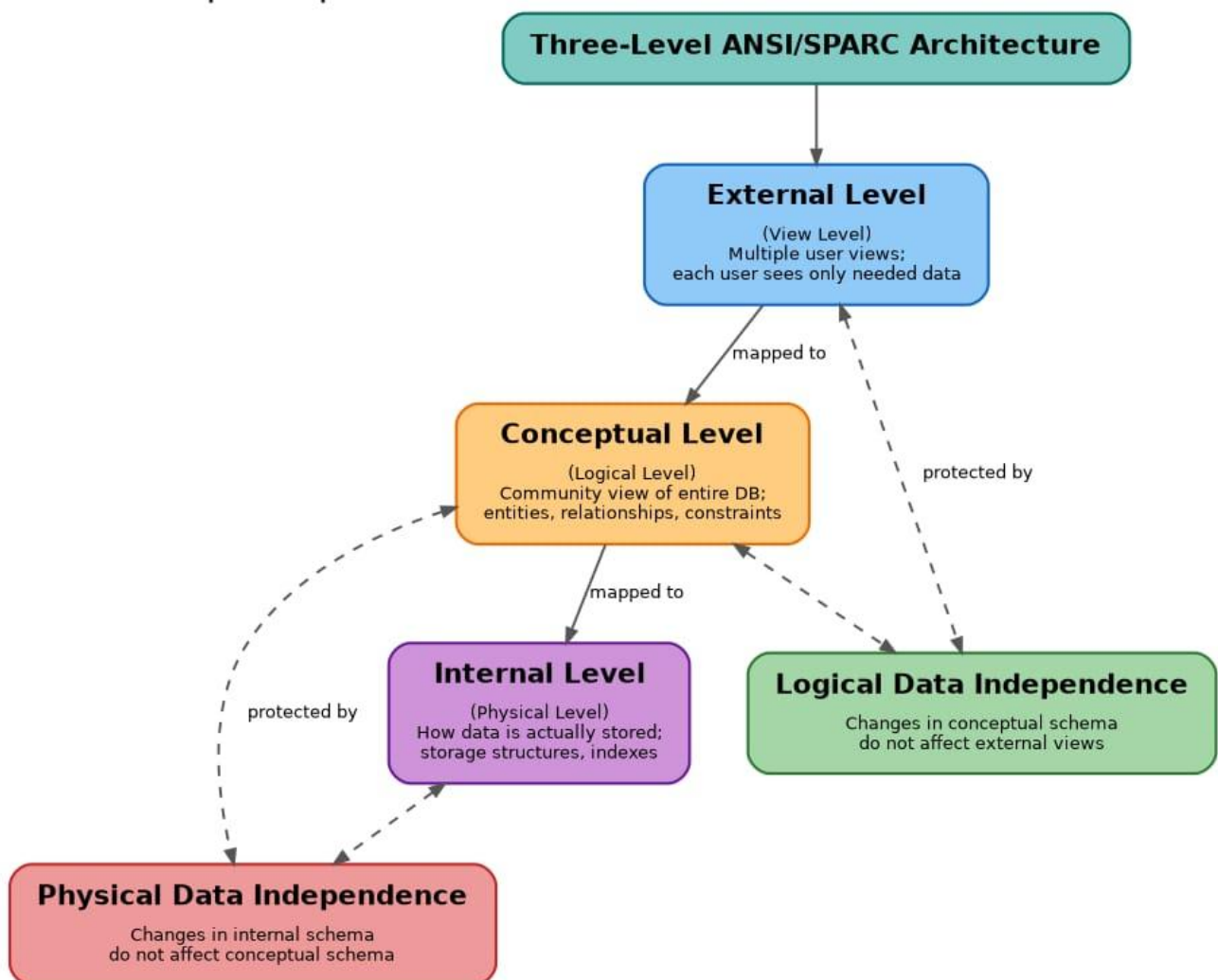
DBMS Core Terms: DBMS uses a Data Model, shown via an ER Diagram made of Entities, Attributes, and Relationships; it defines the Schema and stores the Instance.

Concept Map 1: DBMS Core Terminology

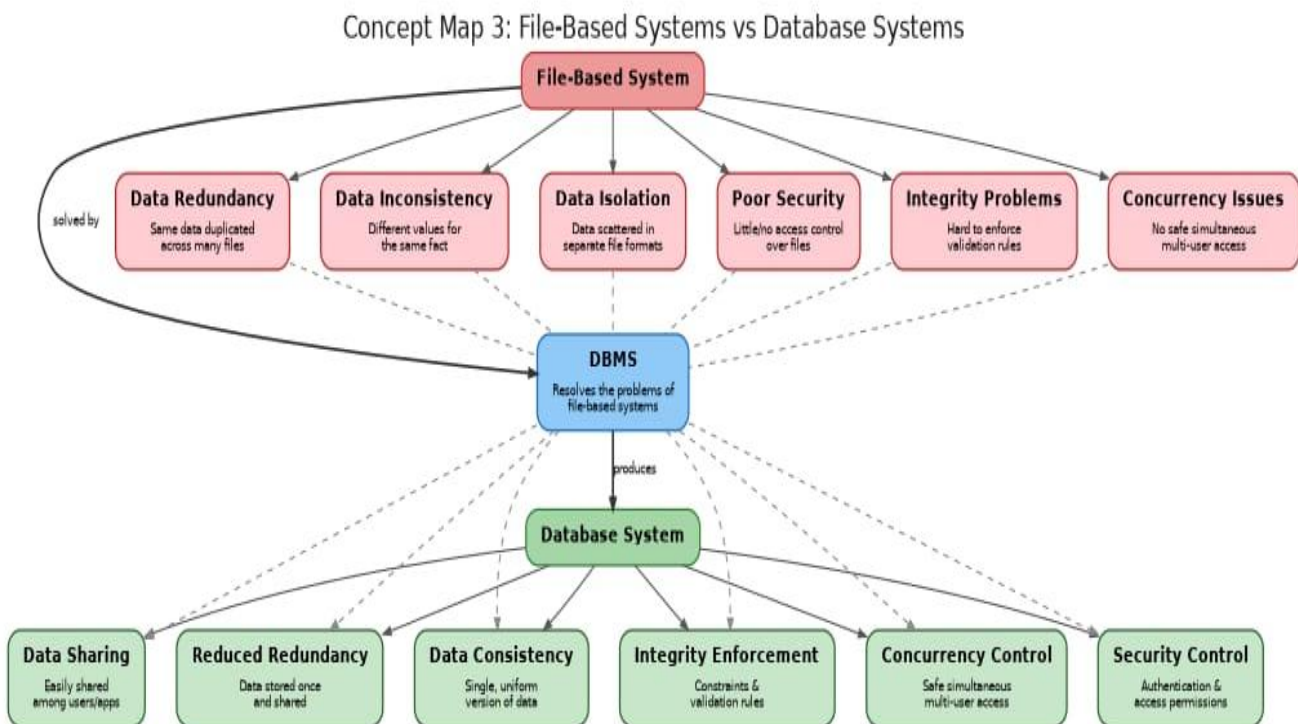


2.ANSI/SPARC: External, Conceptual, and Internal levels, linked by logical and physical data independence.

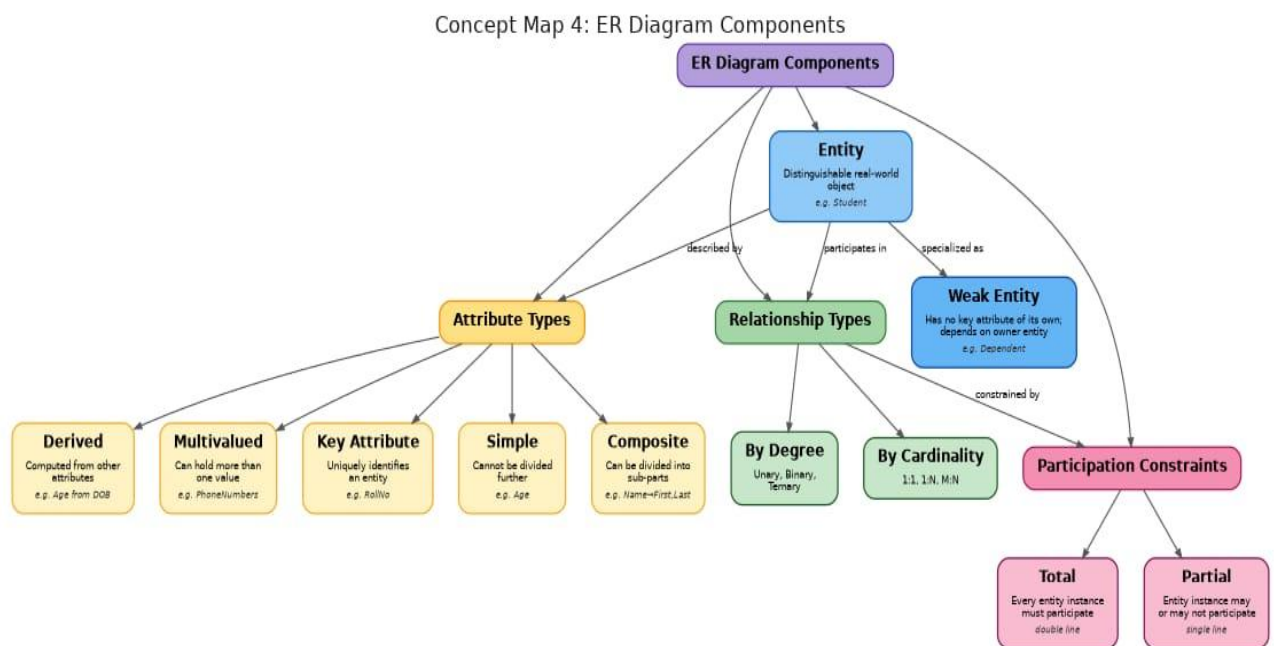
Concept Map 2: ANSI/SPARC Three-Level Architecture



3. File vs Database: File systems have redundancy, inconsistency, and poor security; DBMS fixes these via centralization, consistency, sharing, and security.

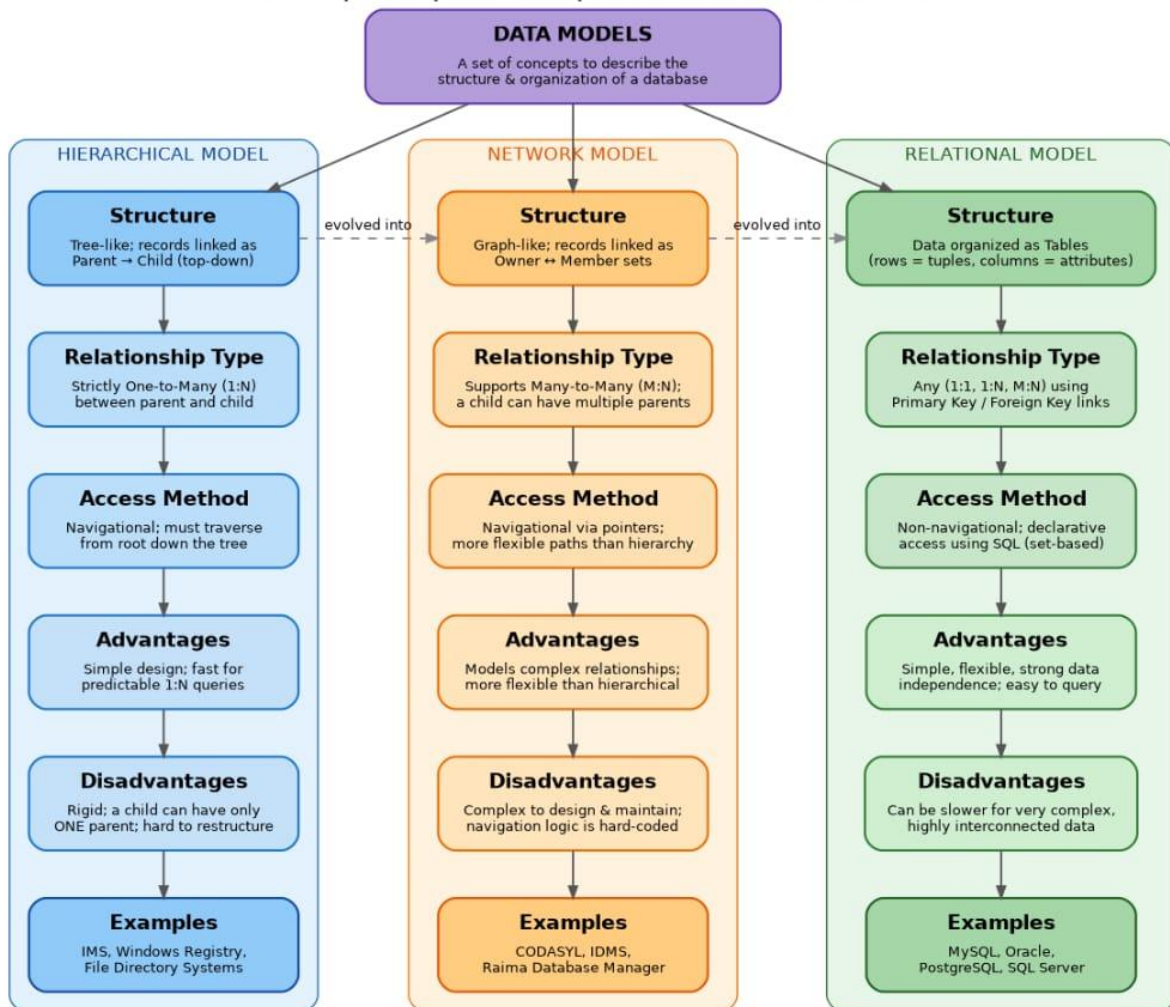


4. ER Components: Entities (strong/weak) have Attributes (simple, composite, derived, multivalued, key); Relationships have degree, cardinality, and participation constraints.



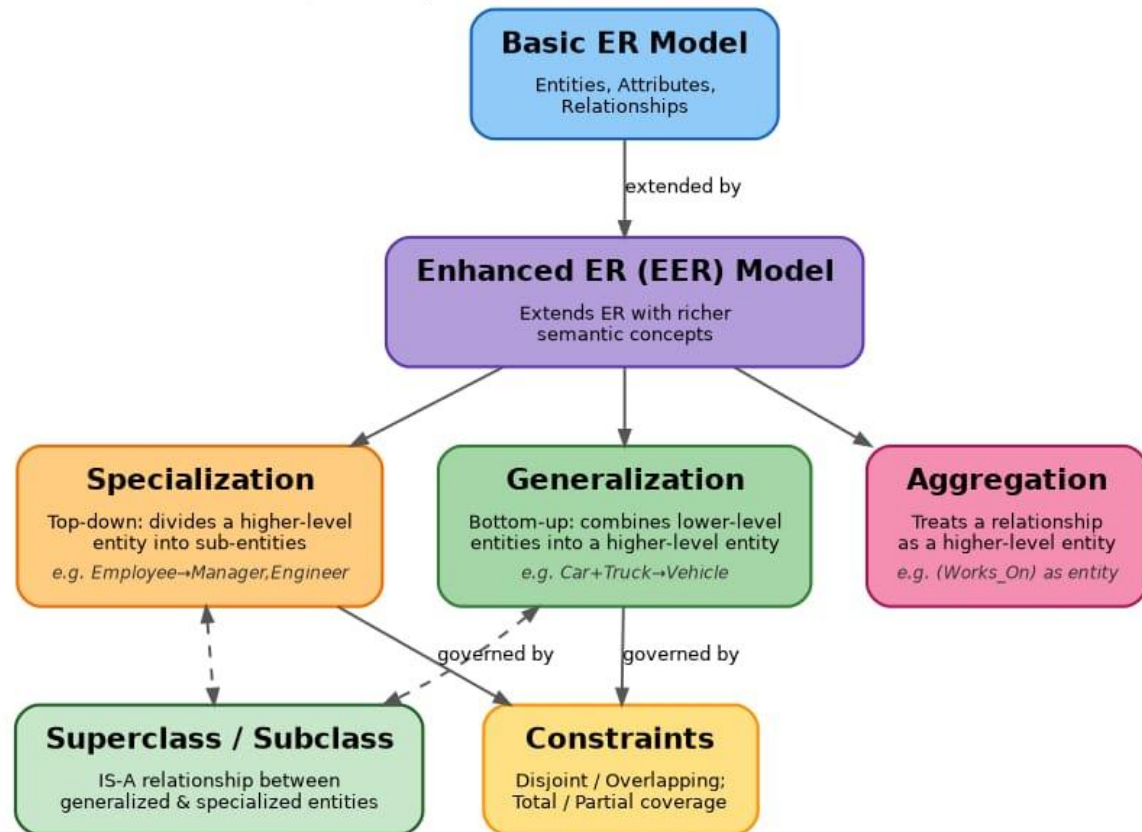
5. Data Model Comparison: Hierarchical (tree), Network (graph), Relational (tables) — Relational evolved from the other two.

Concept Map 5: Comparison of Data Models



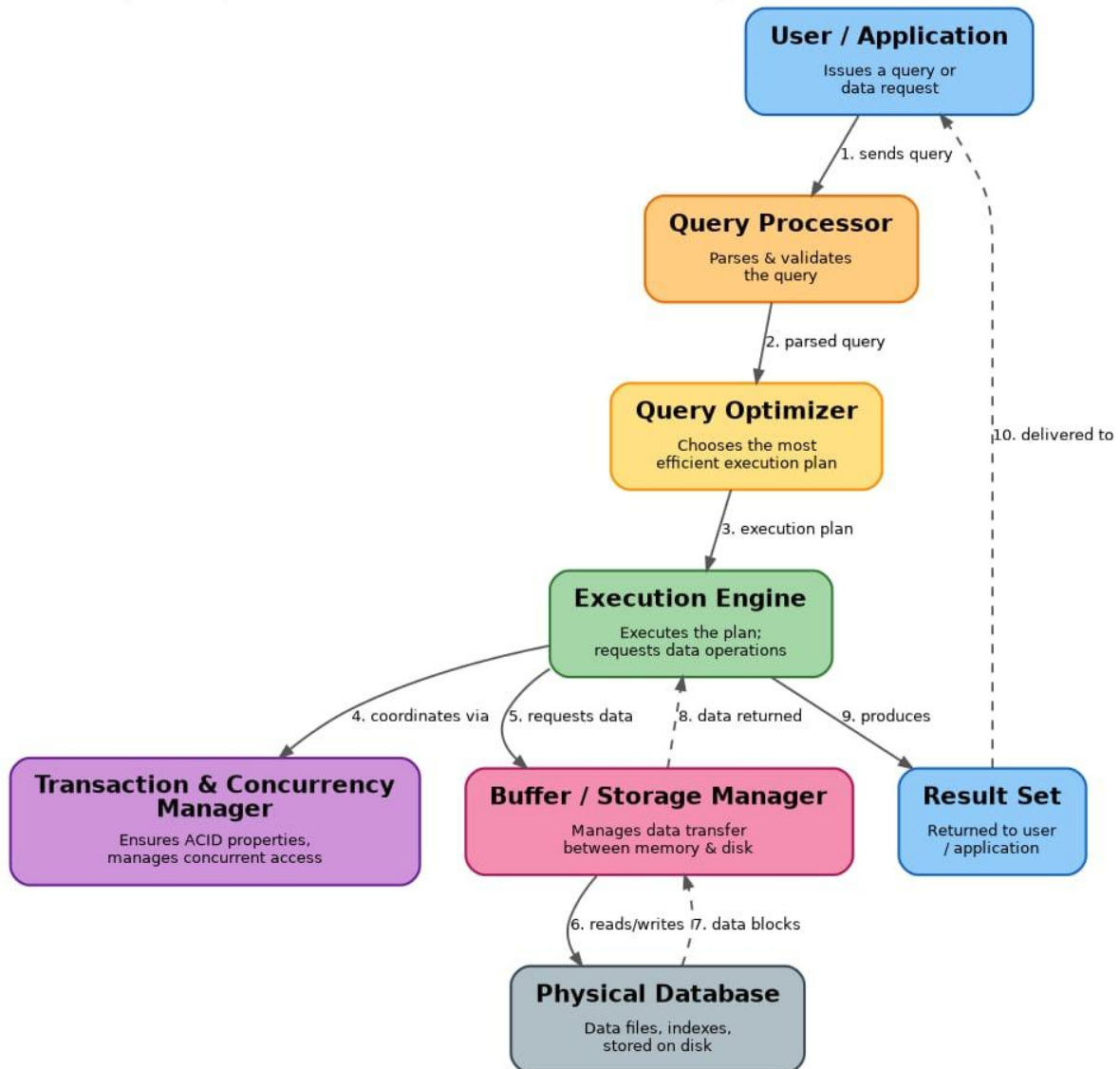
6. EER Model: Generalization (bottom-up), Specialization (top-down), Aggregation (relationship as entity) — governed by disjoint/overlap and total/partial rules.

Concept Map 6: EER Model Extensions



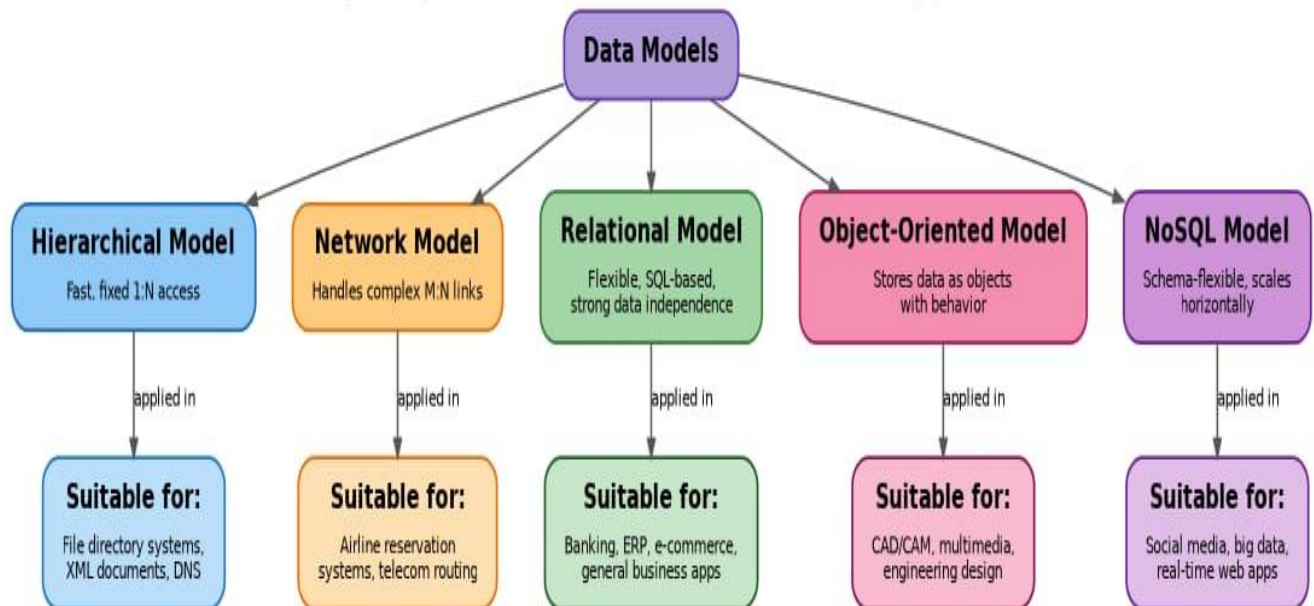
7. DBMS Architecture Flow: Query Processor → Optimizer → Execution Engine → Storage Manager → Physical DB → Result to user.

Concept Map 7: DBMS Architecture - Request to Data Retrieval



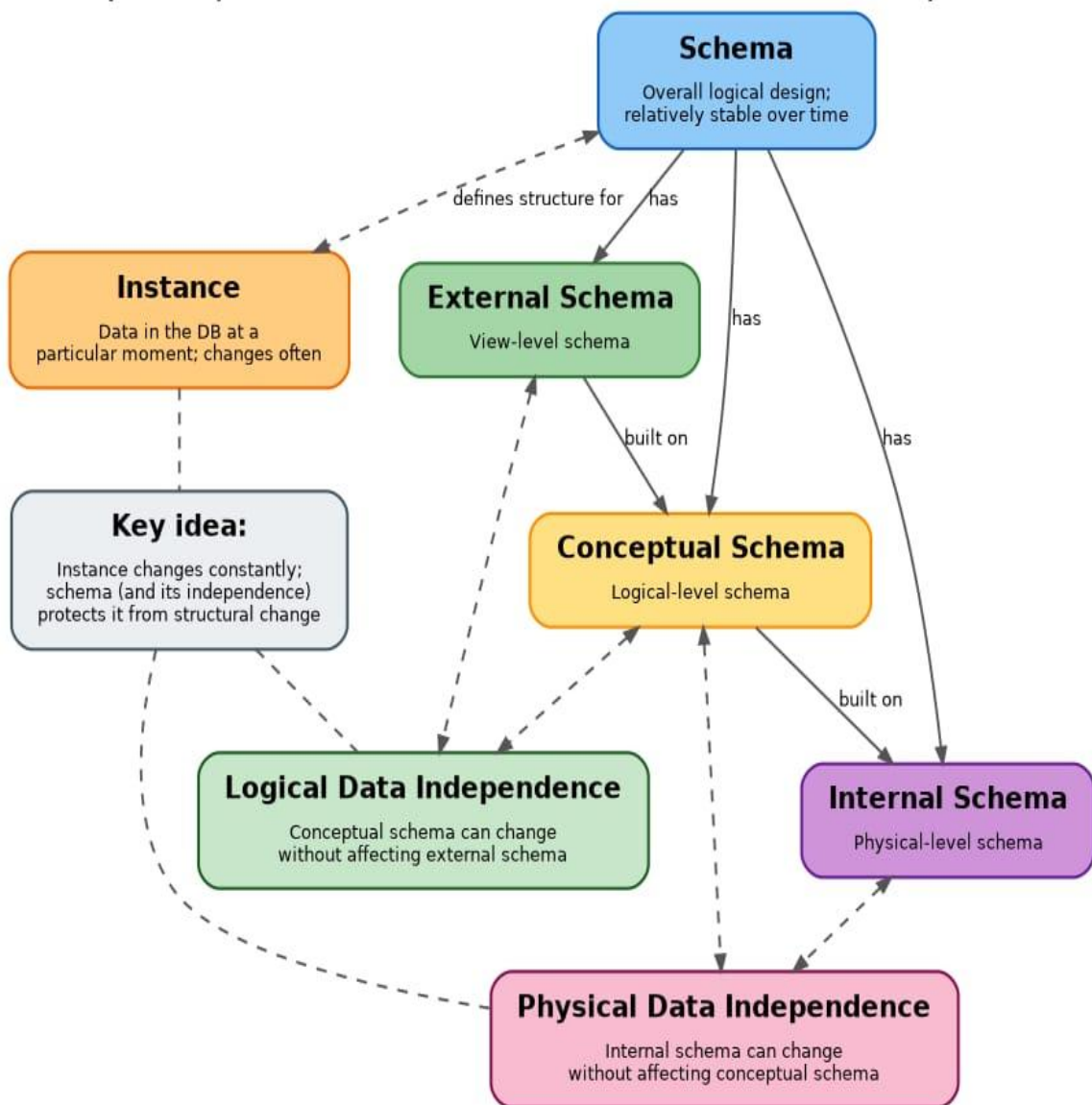
8. Models & Applications: Hierarchical—file systems, Network—airline/telecom, Relational—banking/ERP, OO—CAD/multimedia, NoSQL—social media.

Concept Map 8: Data Models and Real-World Applications



9 . Schema/Instance & Independence: Schema (stable) has three levels; Instance (changing data) is protected by logical and physical independence.

Concept Map 9: Schema, Instance and Data Independence



10. DBMS Roles: DBA (security, design, backup), End Users (naive/casual/sophisticated), Developer (builds interfaces/applications).

Concept Map 10: Roles in a DBMS Environment

